VRMicroscope: Integrating Interactive Microscopy Simulation with Task-State-Aware AI Guidance in Virtual Reality

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Abstract—VRMicroscope integrates interactive microscopy simulation with task-state-aware AI guidance. Its VR laboratory supports specimen handling, observation, focus, illumination and objective adjustment, component inspection, and instructional modules for numerical aperture, spatial frequency, structured illumination, and THz s-SNOM. Authored visual mappings and simplified models are distinguished from calibrated measurement. The assistant combines project knowledge, interface state, and available actions; application code validates guidance and supports spatial markers and user-confirmed Chinese/English speech input. In an exploratory study, 18 of 19 participants completed the core workflow (94.7%); mean SUS was 82.4/100 and unweighted NASA-TLX was 32.4/100. A fixed-state gateway regression used 26 cases repeated three times. All 42 operating recommendations were available and 12/12 navigation targets matched; subsequent author review accepted 75/78 final answers (96.2%), with three GC01 responses judged to require clarification under a revised interpretation. Median gateway latency was 1.82 s. These reused-case and single-group results support feasibility and expose a distinction between available actions and accepted guidance; they do not establish general AI accuracy or causal learning gains. The contribution is a shared state representation connecting interaction, bounded simulation, and conversational guidance.

***Index Terms*— virtual reality; microscopy education; interactive simulation; task-state awareness; large language models; speech input.**

I. INTRODUCTION

Microscopy training requires learners not only to identify three-dimensional instrument components, but also to place specimens, adjust focus and illumination, switch objectives, and explain why the observed field changes. A virtual laboratory can provide a repeatable environment for exploring these operations, but it must make clear which feedback corresponds to real principles and which feedback is designed primarily for instructional purposes.

VRMicroscope combines an experimental layer of objects, controls, feedback, and experiment interfaces with an assistant that maps questions to project knowledge and currently accessible activities. The central challenge is consistency among interface state, simulation boundaries, and recommendations: real confocal operations such as pinhole adjustment or Z-stack acquisition must not be offered when absent, while an experiment entry already visible in a component panel should be used directly.

The main contributions of this work are:

1. We build a microscopy experimental environment that connects specimen manipulation, focus and illumination feedback, structural understanding, and three advanced instructional modules, while explicitly stating the inputs, outputs, and modeling boundaries of each activity.
2. We design an AI guidance architecture that combines project knowledge, current component descriptions, an available-action catalog, response validation, spatial markers, and confirmation-based speech input.
3. We evaluate availability and author-reviewed guidance over 78 real-model requests and report an exploratory study with 19 participants, distinguishing gateway regression results from perceived usability, workload, and untested causal learning gains.

The system supports familiarization and conceptual exploration; real-instrument transfer, quantitative imaging, and incremental learning benefits from AI remain outside the present evidence.

II. RELATED WORK AND RESEARCH POSITIONING

Recent immersive-learning assistants connect generative AI to concrete activities. Chheang et al. separately reported task performance, SUS, NASA-TLX, presence, and interviews for VR anatomy assistants [1]. Geris and Alce logged latency, response duration, and cost for voice-based XR tutoring [2], while SpatialTutor combined object awareness, spatial guidance, and LLM support for procedural training [3]. These studies motivate separating interaction performance, task outcomes, subjective experience, and learning effects.

The present work focuses on the correspondence between microscopy learning activities and guidance. An operation must be genuinely available in the current interface, and its resulting feedback must be explained within an explicit scientific boundary of the simulation. We therefore treat the AI as a constrained tutoring layer embedded in the experimental environment rather than as an autonomous instrument-controlling agent, and we report implementation-level constraint behavior separately from user-level usability outcomes.

Quantitative confocal microscopy provides a scientific reference [4], and educational resources inform the NA and spatial-frequency modules [5], [6]. The project nevertheless lacks a calibrated acquisition pipeline, so its graphical mappings are not treated as physical measurements.

III. VR EXPERIMENTAL ENVIRONMENT AND SIMULATION BOUNDARIES

A. Laboratory Organization and Interaction States

The Unity laboratory supports desktop and XR interaction. Roaming, Observing, and Tutorial states coexist with local

component and experiment states; viewpoint, object activation, permissions, and interface visibility determine available activities. Deferred state transitions and separate tutorial checks reduce conflicting input responses. The AI dialog blocks experimental input, so learners close it before acting. Figure 1 connects specimen handling, observation, adjustment, component inspection, and linked experiments in their shared laboratory.

TABLE I
Inputs, outputs, and instructional boundaries of the activities currently implemented in the project.

Activity and player input	Actual output	Intended instructional connection	Modeling boundary
Pick up, place, observe, and remove a specimen	Changes to the runtime specimen's parent object, pose, visibility, and observation environment	Connect specimen placement with instrument observation	Scripted object manipulation; contact mechanics have not been validated
Coarse/fine focusing and objective switching	Visibility, transparency, rendering feedback, and field-of-view size vary with the focus parameter	Search for an observable range first, then refine the adjustment	Manually defined focus ranges; no confocal point-spread-function calculation
Adjust illumination	Changes to light-cone line width and the brightness input to the specimen shader	Compare illumination settings with observation results	Dimensionless display gain, not calibrated illuminance
SuperAssembly and component selection	Exploded layout, component descriptions, and associated experiment buttons	Connect structure, function, and learning activities	Illustrative separation; does not evaluate real mechanical assembly
Adjust numerical aperture	Illustrative changes in light-cone geometry and specimen appearance	Explore collection angle and NA	Fixed-medium instructional model; NA does not uniquely determine magnification
Select spatial frequency and illumination	Grating/diffraction illustration and a Fourier display derived from the specimen texture	Connect periodic structure, diffraction, and spectral support	Simplified structured-illumination explanation
Select an SNOM probe and control the demonstration	Installation, tapping, scanning animations, and illustrative images	Understand the workflow of near-field microscopy	Procedural demonstration using synthetic outputs

B. Specimen Manipulation and Observation

Picking up a specimen creates a tagged runtime copy attached to the hand. Distance and held-object checks gate placement; the program then standardizes pose and scale in the observation environment. Ownership, focus-dependent visibility, and observation mode remain separate. A microscopic camera renders the instrument view to its screen, and objective switching changes that camera's orthographic size.

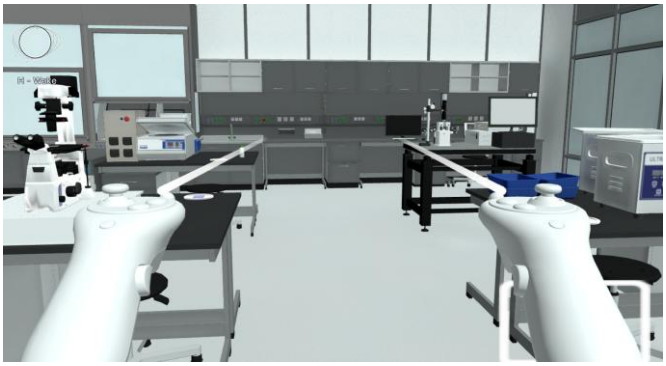
C. Focus, Objectives, and Illumination

The focusing activity uses coarse adjustment to find an observable specimen range and fine adjustment to refine the display. Let v be the dimensionless slider value and j the objective index. For each objective, f_j is the focus center, r_j the visibility half-width, and c_j the full-opacity half-width. With $e_j = |v - f_j|$, the specimen is active within $e_j \le r_j$, with opacity

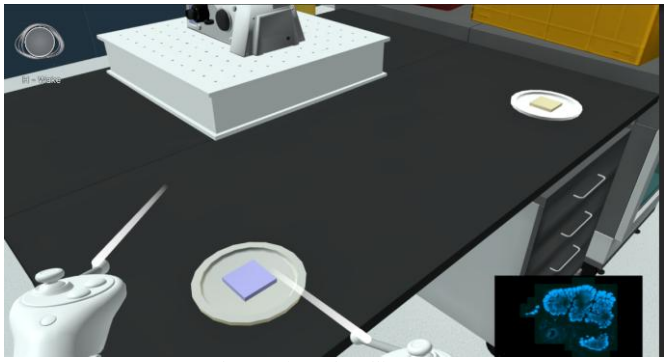
$$\alpha_j(v) = \begin{cases} 1, & e_j < c_j, \\ 1 - \dfrac{e_j - c_j}{r_j - c_j}, & c_j \le e_j \le r_j. \end{cases}$$

Outside this interval the specimen is hidden; within it, a rendering-focus parameter is $300e_j/r_j$. Objective labels 5, 10, 50, and 100 use half-widths 0.17, 0.13, 0.07, and 0.04, with a 25:1 coarse/fine increment ratio. These are visual-feedback parameters, not calibrated displacement or pinhole transmission.

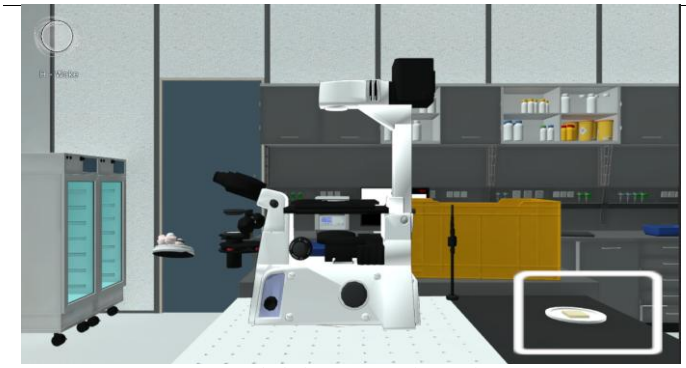
Objective switching rotates the turret and changes viewing scale; illumination changes light-line width and a dimensionless shader input. Focus centers, display gain, and camera sizes are software parameters requiring separate physical calibration.



(a) Laboratory overview with microscope, specimen and assistant entry point.



(b) Specimen handling area.



(c) Exploded component inspection.



(d) SNOM probe-selection station.

Fig. 1. VRMicroscope learning environment and core activities. Learners handle specimens, observe microscope feedback, inspect components, and explore the SNOM workflow within one laboratory, with access to contextual assistant guidance. The panels are runtime captures from the project and are shown as representative views.

D. Structural Exploration and Tutorial Constraints

SuperAssembly separates parts from the model center; learners select expanded components to access names, functions, and linked experiments. The upper optical assembly links to NA and the objective to spatial frequency, with the selected description and usable entry exposed as AI context. Tutorial checks record required operations, not conceptual mastery.

E. Numerical-Aperture Experiment

The NA activity provides a slider range of 0.03–0.95 and uses a fixed refractive index $n = 1$:

$$\mathrm{NA} = n\sin\theta, \qquad \theta = \arcsin(\mathrm{NA}/n).$$

Here θ is the collection-cone half-angle. The interface jointly updates angle and cone geometry, linking numerical input to visible change [5], [7]. Magnification anchors and appearance are illustrative: NA remains distinct from magnification, and changes are not quantitative brightness, depth-of-field, or resolution measurements.

F. Spatial Frequency and Structured-Illumination Displays

The spatial-frequency interface offers gratings of 250, 125, and 62.5 lines/mm and illumination choices. For grating frequency ν_g, illustrative wavelength λ, and illustrative focal length f_{obj}, the diagram uses

$$D = \frac{1}{\nu_g}, \qquad \sin\psi = \frac{\lambda}{D}, \qquad \rho = f_{\mathrm{obj}}\tan\psi.$$

The symbol ρ denotes the back-focal-plane offset, distinct from the specimen spectrum below. At small angles, $\rho/f_{\mathrm{obj}} \simeq \lambda/D$, linking grating spacing to diffraction position [6], [8], [9].

A specimen texture becomes a scalar field; a separable Hann window reduces boundary effects [10], $(-1)^{x+y}$ centers zero frequency, and row/column radix-2 FFTs compute the spectrum [11]. The default is 128×128 (supported: 64–256 powers of two), followed by log compression and display normalization.

Let $O(\boldsymbol{k})$ be the specimen spectrum, $\boldsymbol{k}_0$ the illumination carrier, μ the modulation depth, and φ the illumination phase. The standard SIM background is summarized by [12], [13]

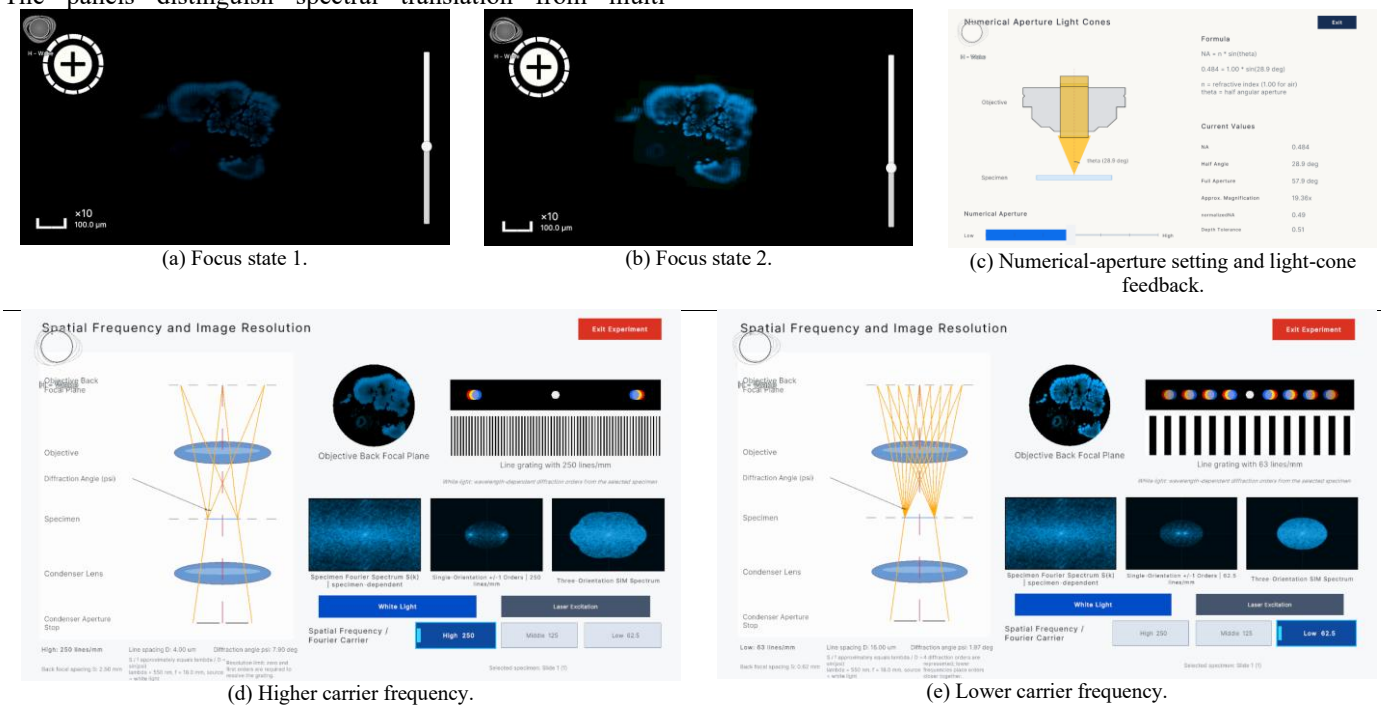
$$G_\varphi(\boldsymbol{k}) = H_{\mathrm{opt}}(\boldsymbol{k})[O(\boldsymbol{k}) + \frac{\mu}{2}e^{i\varphi}O(\boldsymbol{k}-\boldsymbol{k}_0) + \frac{\mu}{2}e^{-i\varphi}O(\boldsymbol{k}+\boldsymbol{k}_0)].$$

Here H_{opt} is the theoretical optical transfer function. The current mixed-spectrum panel illustrates only one shifted sideband pair. Its magnitude before display mapping is

$$B(\boldsymbol{k}) = P_{\mathrm{demo}}(\boldsymbol{k})\left|\frac{\mu}{2}[O(\boldsymbol{k}-\boldsymbol{k}_0) + O(\boldsymbol{k}+\boldsymbol{k}_0)]\right|.$$

P_{demo} is an implementation-defined pupil weight, not a measured H_{opt}; B is further display-mapped. A second panel combines central and shifted coverage at 0°, 60°, and 120°. The panels distinguish spectral translation from multi-orientation coverage without multiphase acquisition, phase separation, or quantitative reconstruction.

The corresponding runtime feedback is summarized in Fig. 2.



(a) Focus state 1.

(b) Focus state 2.

(c) Numerical-aperture setting and light-cone feedback.

(d) Higher carrier frequency.

(e) Lower carrier frequency.

Fig. 2. Mapping instructional inputs to observable feedback. Focus, NA, and illumination carrier affect specimen display, cone geometry, and sideband/support displays, respectively. The feedback combines authored mappings and simplified numerical visualization; the captures identify the changed input rather than claiming calibrated optical measurements.

G. THz s-SNOM Demonstration

The SNOM module covers probe selection, installation, startup, THz generation and propagation, tapping, near-field coupling, background suppression, and raster scanning [14]–[17]. Animation, a scan cursor, synthetic waveforms, and images connect stages to outputs; higher-harmonic displays illustrate separating near-field contributions from background [16], [17]. Probe labels 3×, 2×, and 1× denote relative instructional detail. The module does not demodulate measured signals or perform amplitude/phase inversion or dielectric retrieval.

IV. AI GUIDANCE INTEGRATED WITH THE LABORATORY

A. Conversation Entry and Knowledge Flow

A corner assistant provides local greetings and sends explicit questions through a Python gateway to DeepSeek. The gateway combines versioned project knowledge, interaction descriptions, current snapshots, and a shared action catalog without exposing project files or the rendered view. Speech follows a separate entry path, producing an editable transcript before confirmation invokes the same question-answering pipeline.

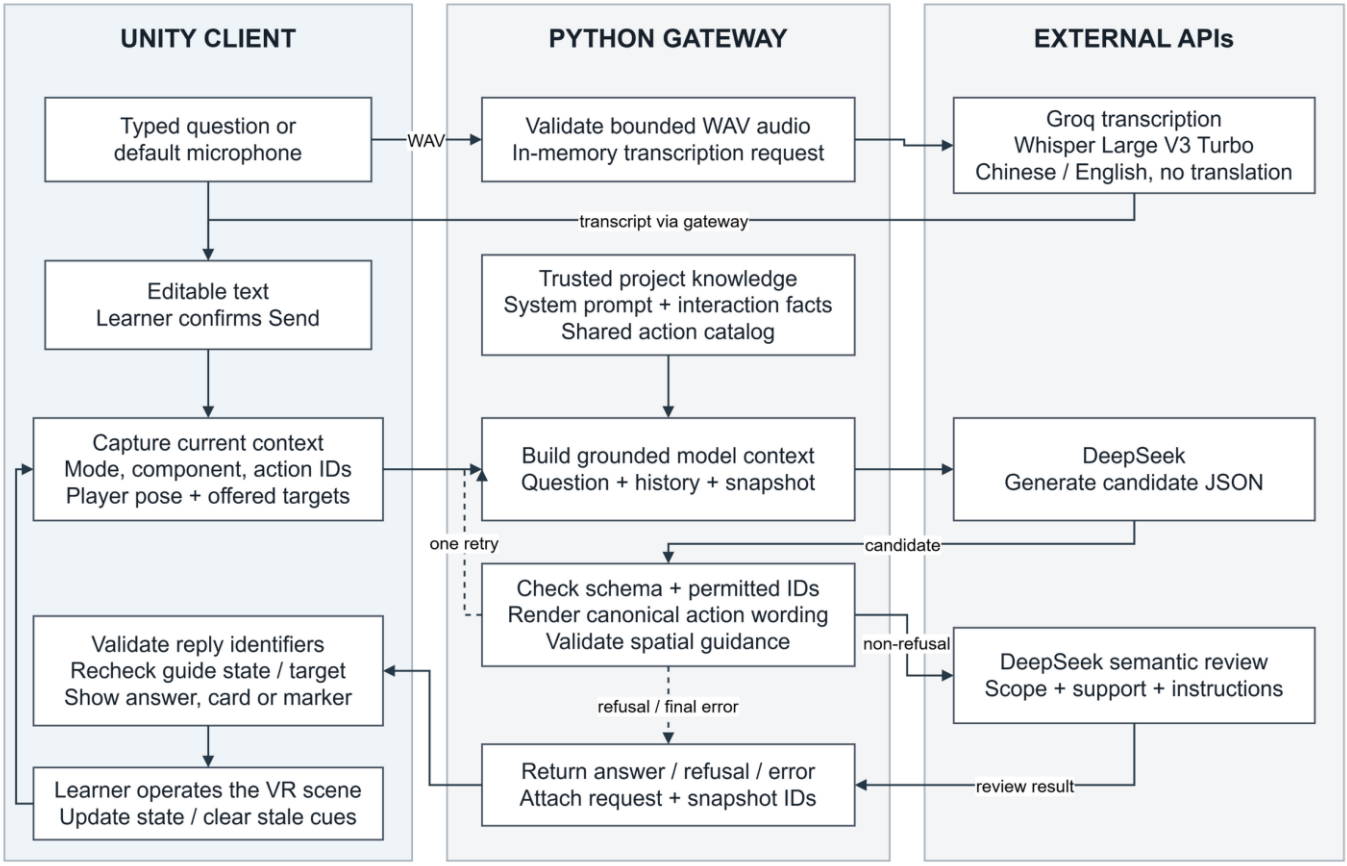


Fig. 3. State-constrained question-answering architecture. The model selects candidate explanations or guidance, while application code checks availability, renders operating instructions, and verifies freshness. Format repair retains the same permissions; learners execute experimental operations.

B. Task State and Available Activities

TABLE II
Correspondence between VR activities and AI context.

VR context	Supplied information	Supported guidance
Specimen handling	Hand object, placement state, available actions	Pickup, placement, observation, or removal
Microscope observation	Mode, observation point, available controls	Focus, brightness, or objective instructions
Component inspection	Name, description, linked experiment, usable start action	Component explanation and experiment entry
NA/spatial frequency	Current module parameters and controls	Parameter interpretation and available steps
SNOM	Stage, probe state, playback, component mode	Workflow and prerequisites
Navigation	Offered targets, positions, player heading, scale	Instrument-area or specific-specimen markers

The snapshot excludes focus values, focus error, objective index, and image quality. Guidance can therefore name a control and observation goal but cannot infer a guaranteed adjustment direction; inactive default fields are not treated as measurements.

C. Constrained Generation and Bounded Repair

For shared catalog $\mathcal{C}$ and action-availability predicate $P(a, s)$ in state s,

$$\mathcal{A}(s) = \{a \in \mathcal{C} \mid P(a, s)\}.$$

Unity constructs $\mathcal{A}(s)$ from controls, controller state, proximity, and bindings. DeepSeek returns an explanation, guidance, clarification, or refusal; operating guidance selects one permitted action and location guidance one offered target. The gateway and client validate correspondence before display.

For valid operations, canonical catalog instructions replace generated prose. Structural failure permits one regeneration from the original question and snapshot with unchanged permissions; timeouts and upstream failures are not treated as format errors.

Non-refusal candidates receive a semantic review for scope, factual support, and goal correspondence. An otherwise supported non-guide answer containing direct control instructions may receive one further regeneration to remove those steps or select a permitted action; the replacement is validated and reviewed again. Generation and review use the same model configuration, so review is an additional check rather than an independent correctness guarantee. Refusals, format errors, access failures, and timeouts remain distinct outcomes.

D. Learning Continuity and Response Freshness

Component references resolve from the selected name and description; hands-on requests prioritize its available experiment, while next-step requests use recent goals and available prerequisites. Post-run review identified ambiguity in a generic next-step question; Section VI-F distinguishes the tested policy from the subsequent clarification revision. Each request stores serialized state and an identifier. Before displaying guidance, the client rechecks identity, state, and permissions; stale recommendations are rejected. Reminders can persist after the dialog closes but never execute or mark an operation complete.

E. Instrument and Specimen Localization

Component, NA, and spatial-frequency topics share the microscope station; SNOM uses a separate apparatus. Red, green, blue, and yellow specimens are distinct targets. Only snapshot-offered objects can be marked, so this is target localization rather than general route planning.

For horizontal target displacement Δ_h and u world units per meter, displayed distance is

$$d = \| \Delta_h \| / u.$$

The signed horizontal angle β is quantized as front ($|\beta| \leq 45°$), back ($|\beta| \geq 135°$), or left/right, and is recomputed by the client. An outline and beacon confirm highlighting. Display distance uses the bounds center; arrival uses the nearest bounds point with a default 1.2-m threshold. Approach, cancellation, expiry, or incompatible state clears the marker.

F. User-Confirmed Chinese/English Speech Input

The speech option records up to 30 s from the default microphone, converts samples to mono PCM16 WAV (preferably 16 kHz), and requests Groq whisper-large-v3-turbo transcription [18]. No translation is requested. The editable transcript requires explicit submission and then follows the same state-validation pipeline. Accuracy and latency remain future evaluation targets.

V. INTEGRATED LEARNING SCENARIOS

A. Specimen Observation and Focus Guidance under Limited State Access

The learner places a specimen, enters observation, switches objectives, and searches with coarse/fine focus. The assistant can select an available adjustment and state what to observe, but cannot read focus error, diagnose a black field, or calculate direction from image data absent from its snapshot.

B. Moving from a Component Question into an Experiment

In SuperAssembly, selecting the upper optical assembly exposes its NA description and launch action; after the assistant explains the relationship, the learner closes the dialog, enters the experiment, and compares light-cone changes. The objective similarly links to spatial frequency. From elsewhere in the room, guidance marks the shared microscope station rather than inventing a separate bench (Fig. 4).

C. SNOM Prerequisites and Response Changes

In SNOM, the assistant can suggest only currently permitted probe, installation, wait, exit, or startup actions. Installation completion unlocks startup, followed by tapping, coupling, and scanning controls. The client rejects any returned operation that no longer matches state; this demonstrates procedural coupling, not adversarial robustness.

VI. EVALUATION METHOD AND RESULTS

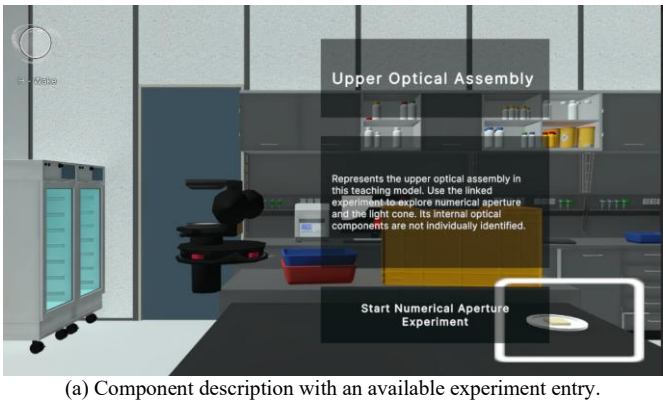
Evaluation separates deterministic implementation conformance from an exploratory user study of workflow completion, usability, and workload, avoiding substitution among code behavior, interaction outcomes, and learning effects [1].

A. Evaluation Questions and Implementation-Level Checks

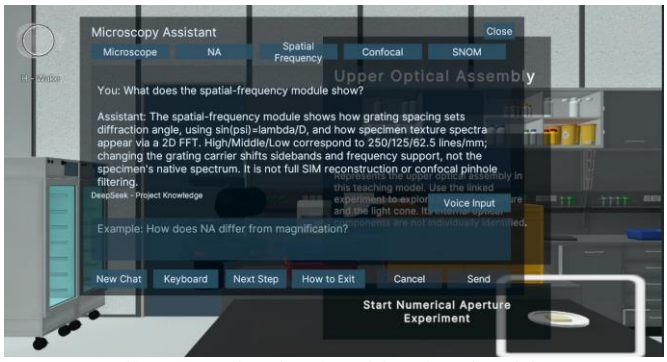
For reporting, we organize the user study around two evaluation questions:

- **RQ1 (workflow feasibility):** Can participants complete the core VR activities involving specimen handling, microscope observation, focus/illumination adjustment, and structural exploration?
- **RQ2 (perceived experience):** How do participants rate the system's overall usability and subjective workload?

Implementation checks concern action-set construction, rejection of absent or unavailable identifiers, stale-response guards, and geometric calculations. Section VI-F additionally measures model/gateway responses and latency under fixed snapshots; that replay does not execute live-client freshness or marker behavior. Deterministic conformance is not reported as AI accuracy, and rendering and speech performance remain outside the present measurements.



(a) Component description with an available experiment entry.



(b) Contextual explanation of the selected module.



(c) Spatial guidance with a target marker.



(d) Speech-input interface.

Fig. 4. Representative assistant interfaces: (a) component description and experiment entry; (b) contextual explanation; (c) spatial target guidance; and (d) speech input. These views illustrate interface states, not one uninterrupted session. The associated NA experiment feedback is shown in Fig. 2(c).

B. Participants and Procedure

The study included 19 participants; academic background describes sample composition rather than experimental groups or optics expertise.

TABLE III
Academic background of participants (N = 19).

Academic background	**n**
Physics	6
Software Engineering	10
Business	2
Philosophy	1

Quiet indoor sessions used a Meta Quest 3. After about three minutes of controller familiarization, participants handled and placed a specimen, entered observation, adjusted coarse/fine focus, switched objectives, adjusted illumination, explored SuperAssembly, and optionally consulted the AI. AI use was not a separate condition, so the study cannot identify an AI-versus-no-AI effect.

Participants then completed SUS [19], NASA-TLX [20], and an approximately 10-minute semi-structured interview. Quantitative analysis uses completion and questionnaires; interviews provide formative feedback and were not coded as qualitative outcomes.

C. Measures and Analysis

Completion is the proportion finishing the required sequence. SUS measures perceived usability [19]; NASA-TLX measures workload using the unweighted six-subscale Raw TLX [20], with lower Performance scores denoting greater perceived success. Measures are descriptive, and SUS is not treated as a knowledge score [21], [22].

D. User-Study Results

TABLE IV
Descriptive outcomes for the full sample (N = 19).

Measure	**Result**
Core-task completion	18/19 (94.7%)
SUS (0–100)	82.4
NASA-TLX: Mental Demand (0–100)	42.9
NASA-TLX: Physical Demand (0–100)	27.9
NASA-TLX: Temporal Demand (0–100)	30.8
NASA-TLX: Performance (0–100)*	23.7
NASA-TLX: Effort (0–100)	44.7
NASA-TLX: Frustration (0–100)	24.2
NASA-TLX: Overall Workload (R-TLX, 0–100)	32.4

**Note: NASA-TLX Performance is scored on a workload-oriented scale (0 = Good/high success, 100 = Poor/failure) consistent with Hart & Staveland [20]; lower values reflect higher self-rated success. Overall Workload is the unweighted Raw TLX (R-TLX) composite score across all six subscales.*

Eighteen participants completed the sequence; SUS of 82.4 indicates high perceived usability [22]. Effort and Mental Demand were the largest workload dimensions. Without a control condition, these differences remain descriptive.

E. Interpretation and Validity Boundaries

This single-group study supports feasibility and perceived usability, not causal AI learning gains or real-instrument transfer. Uneven academic categories are not compared statistically. Speech, frame rate, and adversarial robustness need separate tests; Section VI-F evaluates model/gateway responses.

F. Fixed-State AI Guidance Benchmark

We replayed 26 implementation-derived cases three times through the gateway (78 requests): eight availability, eight goal, six boundary, and four navigation cases. Desktop snapshots and synthetic geometry were supplied, without live Unity capture. Goal-relevant actions satisfy $\mathcal{G}(s,q) \subseteq \mathcal{A}(s)$. The original rubric required operations in 14 cases (42 requests) and navigation in four (12 requests).

The 19 September 2026 run used DeepSeek deepseek-flash, protocol assistant-chat-v4-guidance, knowledge 0a594b7b56cbda74, temperature 0.2, 1,500 output tokens, disabled thinking, non-streaming JSON, and an 18-s timeout. Seed 2027 controlled request order only; the model alias is not an immutable revision.

All 42 raw/final operating actions were available and matched the original goal labels; all 12 navigation targets matched. Subsequent author review of Chinese-translated question–answer pairs accepted 75/78 final responses (96.2%). This single binary review had no independent raters; operating answers used canonical catalog prose.

TABLE V
Gateway replay and subsequent manual review. Operation decisions retain the original 42-request cohort.

Metric	Result
Completed / initial contract valid	78/78 each
Available actions, raw / final	42/42 each
Operation decisions, manual	39/42 (92.9%)
Navigation targets, raw / final	12/12 each
Accepted answers, manual	75/78 (96.2%)
GC01 clarification failures	3/3
Internal review accepted	66/66
Structural / semantic repairs	0/78 / 0/78
Latency, median / P95	1.82 s / 2.30 s

All three failures were GC01: “What should I do next?” elicited expansion instructions in PreAssembly. The request included a prior NA-learning goal and its upper-optical-assembly link, but the review sheet omitted this history. The original rubric rewarded expansion; the reviewer required clarification. This is a post-run interpretation and rubric revision, not an availability violation. Retaining GC01 in the original operation cohort gives 39/42 accepted decisions. All three replies passed internal semantic review, demonstrating that acceptance by this gate does not establish human agreement.

Refusals skipped review: 78 generations and 66 reviews produced 144 provider calls, with no timeout or repair. Latency includes dispatch, generation, review, network, and editor processing, excluding speech and visible rendering; P95 uses nearest rank.

The prompt and GC01 expectation were subsequently revised to clarify; no revised-model result is reported. These 78 requests repeat 26 development cases, with post-hoc author judgments. Translation and omitted review context warrant independent full-context review. Held-out generalization, live-client display/markers, stale-response rejection, and state-ablated comparison remain untested.

VII. DISCUSSION AND LIMITATIONS

Shared state connects components, controls, feedback, and guidance. GC01 illustrates that an available action may still be considered premature. Neither deterministic validation nor same-model review establishes semantic correctness.

The authored mappings and synthetic imagery support instruction, not a calibrated digital twin. Scientific fidelity needs domain-expert review and comparison with real instruments.

The single-group study cannot isolate tutoring effects [1], [21]. Controlled AI-on/off comparisons, knowledge pre/post tests, and expert-scored transfer tasks remain necessary.

VIII. CONCLUSION AND FUTURE WORK

VRMicroscope connects microscopy interaction, bounded simulation, and state-aware guidance. The 19-participant study supports feasibility and perceived usability. Author review accepted 75/78 benchmark answers (96.2%), despite all 42 operating actions being available; GC01 exposed disagreement over clarification. Median latency was 1.82 s. Future work will evaluate the revised policy using held-out cases, independent full-context ratings, live-client tests, and controlled learning/transfer studies.

RESEARCH ETHICS AND GENERATIVE AI DISCLOSURE

This study involved an exploratory usability evaluation of educational virtual reality software with adult participants. No formal institutional ethics review was sought or obtained. The research team judged the activity to be a minimal-risk educational software usability evaluation collecting anonymized questionnaires and formative feedback; this was an author judgment, not an institutional exemption determination. Before participation, individuals were informed of the study objectives and data-collection procedures and provided informed consent. Participation was voluntary, and participants could withdraw at any time without penalty.

Codex assisted with implementation and manuscript content revision, and consistency checks; DeepSeek supplies runtime question answering. Human authors are responsible for verifying the implementation description, study records, analyses, and references.

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